

C Programming Recursion

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What is recursion?

- The process in which a function calls itself directly or indirectly is called recursion and the corresponding function is called as recursive function

Reference: <https://www.geeksforgeeks.org/recursion>

Tips: factorial function

In this program, the factorial function calls itself recursively to calculate the factorial. When n becomes 0 (base case), the function returns 1. Otherwise, the function returns the product of n and $\text{factorial}(n - 1)$ (recursive step).

```
//Another example
long long factorial(int n) {
    if (n > 1) {
        return n * factorial(n - 1);
    }
    else {
        return 1;
    }
}
```

```
#include <stdio.h>

// 関数のプロトタイプ宣言
long long factorial(int n);

int main() {
    int num;
    printf("整数を入力してください: ");
    scanf("%d", &num);

    // 階乗を計算
    printf("%d の階乗は %lld です。%n", num, factorial(num));

    return 0;
}

// 再帰を用いて階乗を計算する関数
long long factorial(int n) {
    // ベースケース
    if (n == 0) {
        return 1;
    }
    // 再帰ステップ
    else {
        return n * factorial(n - 1);
    }
}
```

Pre-learning

- If you haven't seen the video below, please do so before working Exercises.
 - https://wsdmoodle.waseda.jp/mod/millvi/view.php?id=5822904&user_id=26514&courseid=321864
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Recursion (1) 00:24:11



Recursion (2) 00:32:51

Exercise 1, 2

Implement the following program using recursion:

1. Program for finding the greatest common divisor of an integer (a, b) given by a keyboard
 - Integer a and integer b assign values by keyboard input
2. A program to find the power (n) of an integer (x) given by the keyboard
 - Integer x and multiplier n assign values by keyboard input

Exercise 3

- **Create a program that calculates nCr in a form that eliminates recursion.**
 - Calculations such as $100C5$ may overflow, so it is better to define the variables that substitute the calculation results as long long.
 - Example: long long result;
- **Measure the execution speed of the recursive nCr calculation and the recursive-removed nCr calculation, and see which one is more efficient**
 - Try with some numbers
 - Please output the result in a form that shows the measured speed.

Please create the above programs separately.

Tips

How to calculation of combination(nCr)

$${}_nC_r = \frac{{}_nP_r}{r!} = \frac{n(n-1)\cdots(n-r+1)}{r(r-1)\cdots 2\cdot 1} = \frac{n!}{r!(n-r)!}$$

$$\begin{aligned} {}_5C_3 &= \frac{5\cdot 4\cdot 3}{3\cdot 2\cdot 1} \\ &= 10 \end{aligned}$$

The “choose one element” case and the “don't choose one element” case. In each case, recursively find the combination, and finally add up the results.

$${}_nC_r = {}_{n-1}C_{r-1} + {}_{n-1}C_r$$

Example of a program that measures execution speed

```
#include <stdio.h>
#include <time.h>
int main(void)
{
    clock_t start_time, end_time;
    start_time = clock();

    //Write the process you want to measure here

    end_time = clock();
    printf("time: %f¥n", (double)(end_time-start_time));
    #printf("time: %f¥n", (double)(end_time-start_time)/CLOCKS_PER_SEC);

    return (0);
}
```


Exercise 4

Tower of Hanoi

- Create a program to solve the Tower of Hanoi using recursion.
- Please refer to the slide on the next page for the rule.

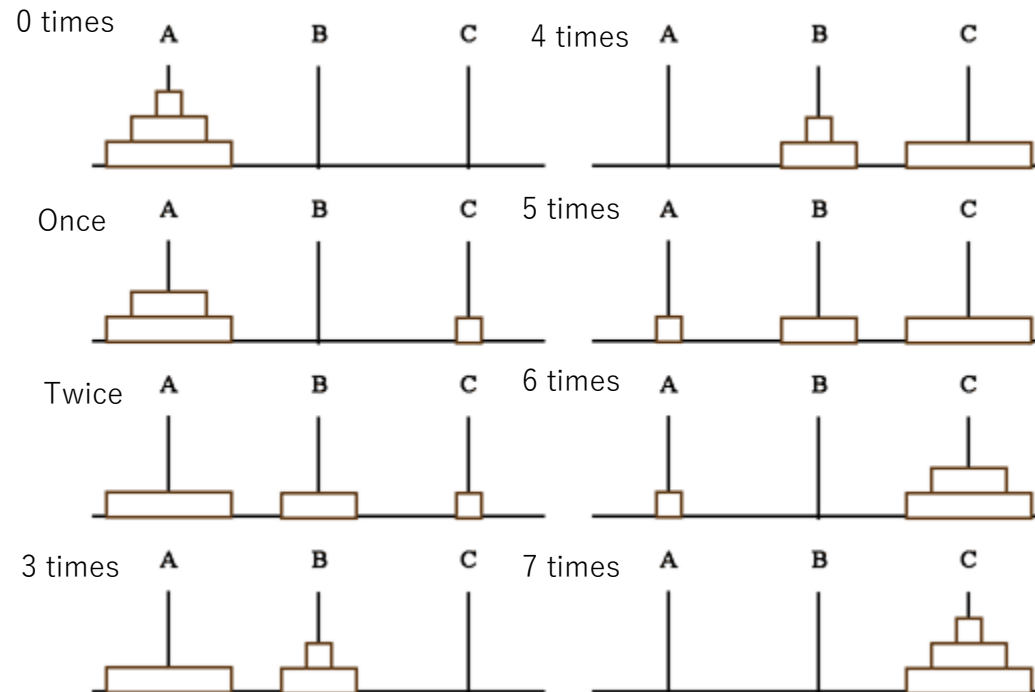


Image Reference : <http://www13.plala.or.jp/kymats/study/C++/Hanoi/Hanoi.html>

Exercise 4

Tower of Hanoi

Tower of Hanoi is a mathematical puzzle where we have three rods (A, B, and C) and N disks. Initially, all the disks are stacked in decreasing value of diameter i.e., the smallest disk is placed on the top and they are on rod A. The objective of the puzzle is to move the entire stack to another rod (here considered C), obeying the following simple rules:

1. Only one disk can be moved at a time.
2. Each move consists of taking the upper disk from one of the stacks and placing it on top of another stack i.e. a disk can only be moved if it is the uppermost disk on a stack.
3. No disk may be placed on top of a smaller disk.

Reference : <https://www.geeksforgeeks.org/c-program-for-tower-of-hanoi/>

About Submitting Exercises

- Summarize the exercises you worked on in Word.
 - Source code
 - Screenshot of execution result
 - Program Description
- Submission is not mandatory. However, these exercises need to be summarized in a report at Lecture 9:Report 2. Therefore, please be sure to work towards Lecture 9..